

MATHEMATICAL ANALYSIS 2

Improper integrals (supplementary exercises with solutions)

 **Problem 1.** Compute $\int_0^\infty \frac{x^2+1}{x^4+1} dx$

 **Solution.**

Standard method: partial fractions decomposition.

Alternative method:

$$\frac{x^2+1}{x^4+1} = \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} = \frac{\left(x - \frac{1}{x}\right)'}{\left(x - \frac{1}{x}\right)^2 + 2}.$$

and let $t = x - \frac{1}{x}$. If $x \rightarrow 0$ ($x > 0$), then $t \rightarrow 0 - \frac{1}{+0} = -\infty$. If $x \rightarrow \infty$, then $t \rightarrow \infty - \frac{1}{\infty} = +\infty$.

$$\int_0^\infty \frac{x^2+1}{x^4+1} dx = \int_{-\infty}^\infty \frac{dt}{t^2+2} = \frac{1}{\sqrt{2}} \operatorname{arctg} \frac{t}{\sqrt{2}} \Big|_{-\infty}^\infty = \frac{1}{\sqrt{2}} (\operatorname{arctg}(+\infty) - \operatorname{arctg}(-\infty)) = \frac{1}{\sqrt{2}} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = \boxed{\frac{\pi}{\sqrt{2}}}.$$



 **Problem 2.** Show that $\int_0^\infty \frac{\ln x}{x^2+1} dx$ is convergent and compute its value.

 **Solution.** Write the integral as the sum of $I_1 = \int_0^1 \frac{\ln x}{x^2+1} dx$ and $I_2 = \int_1^\infty \frac{\ln x}{x^2+1} dx$. We study the convergence for each of them.

Start with I_2 , where the function $\frac{\ln x}{x^2+1}$ is positive on $[1, \infty)$ and use that

$$\lim_{x \rightarrow \infty} x^\alpha \ln x = \begin{cases} 0, & \text{if } \alpha < 0 \\ \infty, & \text{if } \alpha \geq 0 \end{cases}. \quad (1)$$

Then

$$L = \lim_{x \rightarrow \infty} x^p \cdot \frac{\ln x}{x^2+1} = \lim_{x \rightarrow \infty} \frac{x^2}{x^2+1} \cdot \lim_{x \rightarrow \infty} x^{p-2} \ln x = 1 \cdot \begin{cases} 0, & \text{if } p-2 < 0 \\ \infty, & \text{if } p-2 \geq 0 \end{cases} = \begin{cases} 0, & \text{if } p < 2 \\ \infty, & \text{if } p \geq 2 \end{cases}.$$

Use the **C1** convergence test and let $p \in (1, 2)$ (e.g., $p := 3/2$). Then

$$L = 0, \quad p > 1 \Rightarrow I_2 \text{ is convergent.}$$

For I_1 , let $x = \frac{1}{t}$. Then

$$I_1 = \int_0^1 \frac{\ln x}{x^2+1} dx = \int_\infty^1 \frac{\ln \frac{1}{t}}{\frac{1}{t^2}+1} \cdot \left(-\frac{1}{t^2} \right) dt = \int_1^\infty \frac{\ln \frac{1}{t}}{1+t^2} dt = - \int_1^\infty \frac{\ln t}{1+t^2} dt = -I_2.$$

Since I_2 is convergent, it follows that I_1 is also convergent, so the initial integral is convergent. Moreover,

$$I = I_1 + I_2 = 0.$$



 **Problem 3.** Show that $\int_0^{\pi/2} \ln(\sin x) dx$ is convergent.

 **Solution.** Since $\ln(\sin x) \leq 0$ for $x \in (0, \frac{\pi}{2}]$, rewrite $\int_0^{\pi/2} \ln(\sin x) dx = - \int_0^{\pi/2} |\ln(\sin x)| dx$ and study the convergence of $\int_0^{\pi/2} |\ln(\sin x)| dx$, using **C2**. We analyze $L = \lim_{\substack{x \rightarrow 0 \\ x > 0}} x^p |\ln(\sin x)|$.

$$\begin{aligned} L &= \lim_{\substack{x \rightarrow 0 \\ x > 0}} \underbrace{\left(\frac{x}{\sin x}\right)^p}_1 \cdot \lim_{\substack{x \rightarrow 0 \\ x > 0}} \underbrace{(\sin x)^p |\ln(\sin x)|}_{\sin x = t \rightarrow 0 \text{ (when } x \rightarrow 0)} = \lim_{\substack{t \rightarrow 0 \\ t > 0}} \underbrace{t^p |\ln t|}_{\ln t < 0 \text{ for } t \text{ close to } 0} = \lim_{\substack{t \rightarrow 0 \\ t > 0}} t^p (-\ln t) \\ &= \lim_{\substack{t \rightarrow 0 \\ t > 0}} \underbrace{t^p \ln \frac{1}{t}}_{\frac{1}{t} = y \rightarrow +\infty} = \lim_{y \rightarrow \infty} y^{-p} \ln y \stackrel{(1)}{=} \begin{cases} 0, & \text{if } -p < 0 \\ \infty, & \text{if } -p \geq 0 \end{cases} = \begin{cases} 0, & \text{if } p > 0 \\ \infty, & \text{if } p \leq 0 \end{cases}. \end{aligned}$$

Let $p \in (0, 1)$ (e.g., $p := 1/2$), hence

$$L = 0, \quad p < 1 \stackrel{(C2)}{\Rightarrow} I \text{ is convergent.}$$



 **Problem 4.** $\int_0^1 \frac{\ln(1-x)}{x} dx$

 **Solution.** Recall that $\int_0^1 \frac{\ln(1+x)}{x} dx = \frac{\pi^2}{12}$ (from Seminar 7). Let $x = t^2$. Then:

$$I = \int_0^1 \frac{\ln(1-x)}{x} dx = \int_0^1 \frac{\ln(1-t^2)}{t^2} \cdot 2t dt = 2 \int_0^1 \frac{\ln(1-t) + \ln(1+t)}{t} dt = 2I + 2 \int_0^1 \frac{\ln(1+t)}{t} dt = 2I + \frac{\pi^2}{6}.$$

Assuming that I is convergent (prove it – independent study), it follows that $I = \boxed{-\frac{\pi^2}{6}}$.

Remark: A negative number is to be expected, since $\ln(1-x) < 0$ for all $x \in (0, 1)$.



 **Problem 5.** Study the convergence of $\int_0^1 \frac{\ln x}{1-x} dx$.

 **Solution.** Both 0 and 1 are singularities. There are no other singularities inside $(0, 1)$. Also, $\ln x < 0$ for $x \in (0, 1)$, so write $\int_0^1 \frac{\ln x}{1-x} dx = - \int_0^1 \frac{|\ln x|}{1-x} dx$.

For the convergence at 0, apply **C2**:

$$L_0 = \lim_{x \searrow 0} x^p \frac{|\ln x|}{1-x} = \lim_{x \searrow 0} \frac{1}{1-x} \cdot \lim_{x \searrow 0} x^p (-\ln x) = - \lim_{x \searrow 0} x^p \ln x \stackrel{(1)}{=} \begin{cases} 0, & \text{if } p > 0 \\ \infty, & \text{if } p \leq 0 \end{cases}.$$

Choose any $p \in (0, 1)$, hence

$$L_0 = 0, \quad p < 1 \stackrel{(C2)}{\Rightarrow} \text{the integral is convergent at 0.}$$

For the convergence at 1, apply **C3**:

$$L_1 = \lim_{x \nearrow 1} (1-x)^p \frac{|\ln x|}{1-x} = \lim_{x \nearrow 1} (1-x)^p \frac{\ln x}{x-1}$$

and use $\lim_{x \rightarrow 1} \frac{\ln x}{x-1} = 1$. By letting $p = 0$, it follows that

$$L_1 = 1, \quad p < 1 \stackrel{(C3)}{\Rightarrow} \text{the integral is convergent (at 1).}$$

Concluding, the integral is convergent. ✓

Problem 6. Compute $\int_0^1 \frac{\ln x}{\sqrt{1-x^2}} dx$.

Solution. Let $x = \sin t$. Then

$$\int_0^1 \frac{\ln x}{\sqrt{1-x^2}} dx = \int_0^{\pi/2} \ln(\sin t) dt = -\frac{\pi \ln 2}{2}.$$

(see Seminar 7). ✓

Problem 7. Compute $\int_0^1 \frac{\arcsin x}{x} dx$.

Solution. The integration by parts leads to:

$$\begin{aligned} \int_0^1 \frac{\arcsin x}{x} dx &= \int_0^1 \arcsin x \cdot \ln' x dx = \arcsin x \cdot \ln x \Big|_0^1 - \int_0^1 \arcsin' x \cdot \ln x dx \\ &= 0 - \lim_{x \rightarrow 0} (\arcsin x \cdot \ln x) - \underbrace{\int_0^1 \frac{\ln x}{\sqrt{1-x^2}} dx}_{-\frac{\pi \ln 2}{2} \text{ (cf. Problem 6)}} = -\underbrace{\lim_{x \rightarrow 0} \frac{\arcsin x}{x}}_1 \cdot \underbrace{\lim_{x \rightarrow 0} x \ln x}_0 + \frac{\pi \ln 2}{2} = \frac{\pi \ln 2}{2}. \end{aligned}$$

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Problem 8. Compute $\int_0^\infty \frac{\ln(1+x^2)}{1+x^2} dx$.

Solution. Let $x = \tan t$, $\arctan x = t$, $\frac{dx}{1+x^2} = dt$. Then:

$$\begin{aligned} \int_0^\infty \frac{\ln(1+x^2)}{1+x^2} dx &= \int_0^{\pi/2} \ln(1+\tan^2 t) dt = \int_0^{\pi/2} \ln\left(1 + \frac{\sin^2 t}{\cos^2 t}\right) dt = \int_0^{\pi/2} \ln\left(\frac{1}{\cos^2 t}\right) dt = -2 \int_0^{\pi/2} \ln(\cos t) dt \\ &= (-2) \cdot \left(-\frac{\pi \ln 2}{2}\right) = \pi \ln 2 \quad (\text{see Seminar 7}) \end{aligned}$$

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Problem 9. Compute $\int_0^\infty \arcsin e^{-x} dx$.

Solution. Let $e^{-x} = t$. Then see Problem 7. ✓